The Impact of Synchronized Visual and Auditory Attention on Human Perception

Lichuan Jiang[1,2], Jiani Zhong[1], Muqing Jian[1], Xuanzhuo Liu[1], Siqi Cai[1,3](✉), and Haizhou Li[1,3]

[1] School of Data Science, The Chinese University of Hong Kong, Shenzhen 518172, Guangdong, People's Republic of China
elesiqi@nus.edu.sg
[2] TUM School of Computation, Information and Technology, Department of Electrical Engineering, Technical University of Munich, Munich, Germany
[3] Machine Learning Lab (MLL), University of Bremen, Bremen, Germany

Abstract. The cocktail party problem shows the remarkable human ability to selectively attend to and recognize one source of auditory input in a noisy environment. However, individuals may struggle to identify a speaker's voice when they are unfamiliar with the speakers and don't have a clear visual focus, which results in less visual information. This raises the question: How can visual information aid in extracting information from a speaker's voice? This study explores how synchronized visual and auditory attention impact human perception in scenarios involving two speakers. Using Tobii Glasses 3 to track participants' eye movements and pupil diameters, combined with questionnaire responses, we explore how these factors influence speech comprehension. Our results demonstrate that participants achieve higher accuracy in speech comprehension when they focus their gaze on the speaker they are listening to, compared to scenarios where visual attention is divided between speakers or where they rely solely on auditory cues. These findings highlight the effectiveness of synchronizing visual and auditory attention in improving the acquisition and processing of information.

Keywords: Multi-modal · Audiovisual Attention · Human Perception · Speech Comprehension · Pupil Diameter

1 Introduction

In a cocktail party, individuals with normal hearing abilities possess the remarkable capacity to separate and focus on specific speech to gather relevant information [1]. However, even individuals with normal hearing find it increasingly

L. Jiang and J. Zhong—These authors contributed equally to this work as co-first authors.
M. Jian and X. Liu—These authors contributed equally to this work as co-second authors.

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025
H. Li et al. (Eds.): ICSR + InnoBiz 2024, LNAI 15170, pp. 41–50, 2025.
https://doi.org/10.1007/978-981-96-1151-5_5

difficult to recognize a speaker's voice when they are unfamiliar with the speakers and don't have a clear visual focus, which results in less visual information. In these scenarios, the potential role of visual cues remains relatively unexplored. This raises the question of how visual information can aid in extracting the target speaker's voice, particularly in complex auditory environments.

The integration of multiple sensory modalities within the human perceptual system, particularly the synchronization of visual and auditory attention, is essential for enhancing information processing capabilities [2,3]. This synergy is particularly valuable where auditory cues alone may be insufficient. Understanding how visual attention compensates for auditory limitations or enhances overall perception holds profound implications for communication strategies, multimedia design, and educational settings [4,5].

Previous studies in visual and auditory modalities have significantly advanced our understanding of attentional processes [6,7]. Perspectives vary, with some emphasizing constraints in concurrent attentional capacities [8], while others highlight potential synergies among sensory modalities [9]. Recent studies have investigated attention's role in "cross-modal correspondences" [10,11], the dynamic interaction between eye movements and linguistic stimuli [12], and the influence of gaze behavior on multisensory integration [2]. Recent research [12] has provided new insights into eye movements related to attended and unattended natural speech sentences. By integrating simultaneously recorded eye-tracking and magnetoencephalographic data with temporal response functions, the study reveals that eye gaze aligns with attended speech.

Despite these advancements, gaps remain in understanding how attention operates across different modalities, especially in complex auditory environments. Quantitatively measuring attention levels and synchronizing visual and auditory attention are crucial for advancing this field. This study aims to bridge these gaps by integrating eye-tracking and facial recognition technologies to explore individual and interactive sensory processing mechanisms. Specifically, we employ Tobii Glasses 3 to precisely collect eye movement and pupil diameter data, alongside facial recognition through video recordings, ensuring accurate and efficient measurement of participants' attention. Using AI-generated materials, we achieve rigorous control and precision in experimental design, thereby enabling sophisticated investigations into multi-modal attention. The contribution of this paper can be mainly summarized as:

1. Development of a comprehensive dataset that includes audiovisual stimuli, eye movement, and pupil diameter data, along with questionnaires to enhance research in visual and auditory attention.
2. Quantitative analysis exploring the effects of synchronized visual and auditory attention on human perception.
3. Contributing linguistic and cultural diversity through research conducted in Chinese, thereby enriching understanding across different cultural contexts.

2 Methods and Materials

2.1 Hypothesis

We hypothesize that the alignment of visual information plays a pivotal role in human auditory attention and speech Comprehension. To validate this hypothesis, we have designed the following experiment.

2.2 Experiment Setup

As illustrated in Fig. 1, we developed a multi-talker speech perception system to evaluate the impact of synchronized visual and auditory attention. The experiment consisted of six groups, each with four video clips, featuring two AI digital avatars simultaneously speaking different dates in Mandarin Chinese from left and right sections. A continuous period during which a participant watches a video clip is referred to as a "trial", with each participant completing a total of 24 trials.

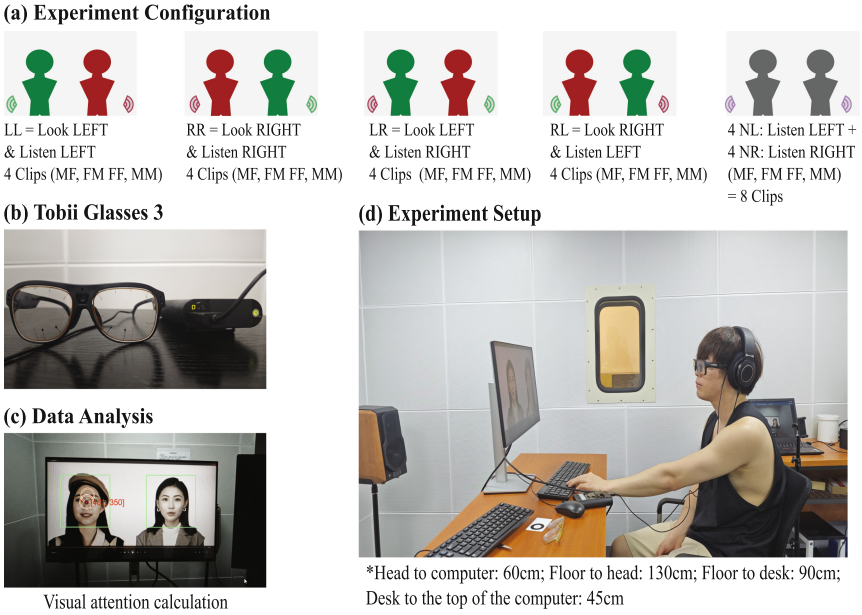

Fig. 1. Experiment overview. (a) Experiment configuration: Two dates are spoken simultaneously by two speakers in the video. Participants are instructed to listen and look toward specified directions. M = Male, F = Female, MF = Male (Left) Female (Right), and others in a similar fashion. (b) Tobii Glasses 3 used for monitoring visual parameters. (c) Data analysis: Visualization of the visual attention calculation procedure. (d) Experiment setup.

To ensure consistent sound input and provide spatial cues, the Head Related Transfer Function (HRTF) [13] was employed. Participants were seated in a soundproof chamber with earphones and eye-tracking glasses. They were instructed to fixate their gaze and listen to the voice according to different condition setups as described below. Subsequently, they recorded the dates they perceived in a questionnaire following each clip.

The experiment was conducted under three conditions: SameSide, DifferSide, and OnlyListen. In the SameSide condition, subjects were instructed to focus their gaze on one side while listening to speech from the same side. In the DifferSide condition, subjects focused their gaze on one side while listening to speech from the opposite side (see 1). The OnlyListen group involved participants listening to speech without any visual cues. Both the SameSide and DifferSide conditions were further categorized into four sub-conditions: RR (both stimuli on the right), LL (both on the left), RL (listening on the right, looking left), and LR (listening on the left, looking right). Participants' gaze directions were monitored using Tobii Pro Glasses 3 [14].

Participants. 16 healthy participants aged between 18 and 22 were recruited from the university. The participant group comprised 6 males and 10 females, including 3 individuals (1 male and 2 females) from Indonesia who were non-native Mandarin speakers. Despite not being native speakers, these three participants exhibited exceptional speech comprehension of the Chinese dates used in our experiment. It should be noted that data from two participants were excluded from the analyses due to failure to follow the experimental instructions.

2.3 Data Acquisition

Eye Movements and Pupil Diameter. Eye movements indicate where visual attention is directed and for how long, showing which speaker captures interest. Meanwhile, changes in pupil diameter reflect the mental effort, offering insights into the cognitive load of tasks [15]. To investigate both visual attention and cognitive effort, we utilized the Tobii Pro Glasses 3 [14], an advanced wearable eye-tracking device that captures real-world visual data while allowing for natural user interactions. This device records visual data from a first-person perspective at a sampling rate of 100 Hz. Specifically, the device records detailed eye movements, pupil diameter changes, and gaze point locations at a sampling rate of 100 Hz. Analysis of these metrics provides valuable insights into the dynamics of visual attention and cognitive workload among participants.

Questionnaire Design. The questionnaire comprises six groups (see Sect. 2.2 for details). Each group contains four questions, where participants are asked to record the dates of the target speaker in each audio clip. The sequence of clips within each group is designed to test the accuracy of identifying males and females under different conditions. The clips are organized as follows:

- One male on the left and one female on the right (MF)
- One female on the left and one male on the right (FM)
- Two males, one on the left and one on the right (FF)
- Two females, one on the left and one on the right (MM)

At the end of each section, participants are asked to provide a subjective rating of their visual attention to the target person, as well as of the attention level of listening (rated from "not focused" 0 to "very focused" 6).

3 Data Analysis and Results

We quantify the visual and audio attention across multiple dimensions:

- Audio Attention: Accuracy of questionnaire responses serves as an indicator.
- Visual Attention: Eye movement and gaze point locations were analyzed to measure visual attention.
- Cognitive workload: Pupil diameter serves as a metric for measuring cognitive workload.

3.1 Analysis of Eye-Tracking Signals

Gaze Point Analysis. Using the face recognition function of OpenCV [16], we calculate the period where video frames occur, in addition to utilizing the timestamp event markers sent by E-Prime. This ensures us a more exact period in which subjects see two faces on the screen. After subtracting the target faces in the video, we compare the location of the faces with the gaze points locations and calculate the level of visual attention V_a using the formula

$$V_a = \frac{number\ of\ \text{`}in - box\text{'}\ frames}{total\ number\ of\ speaker\ frames}$$

Here, the 'box' denotes the region around the target speaker's face determined by computer vision methods [16]. Correct fixation is determined by counting the frames where the gaze point falls within this specified region in each trial. The ratio of correct fixation frames to total frames with faces is computed to provide a measure of attention time proportion.

Pupil Diameter Analysis. Pupil diameter is a significant indicator of a subject's attention level and cognitive load [17–19]. Larger pupil dilations are generally associated with higher cognitive load and concentration [20]. Furthermore, pupil dilation patterns can index information processing and attentional states, with systematic variations corresponding to engagement levels [21]. In this study, we analyzed pupil diameter variability to assess attentional states across different experimental conditions. Specifically, we compared pupil size variability between the SameSide condition (RR and LL) and the DifferSide condition (RL and LR). Pupil size variability, measured as the variance in pupil diameter, serves as an

indicator of cognitive load [22]. Larger variability typically reflects greater pupil dilation.

As shown in Fig. 2(a), the average variability was lower in the SameSide condition compared to the DifferSide condition, with the DifferSide condition exhibiting greater variability. Figure (2(b) provides a detailed view of pupil size variability across participants in different conditions. Although there are individual differences in pupil size variability, the DifferSide condition generally exhibited higher variability across participants. This finding suggests that the synchronized visual and auditory information influences pupil size variability. Specifically, the DifferSide condition likely induces higher cognitive load, emphasizing the critical role of synchronized visual information from the attended speaker's face in optimizing attentional focus and enhancing speech comprehension.

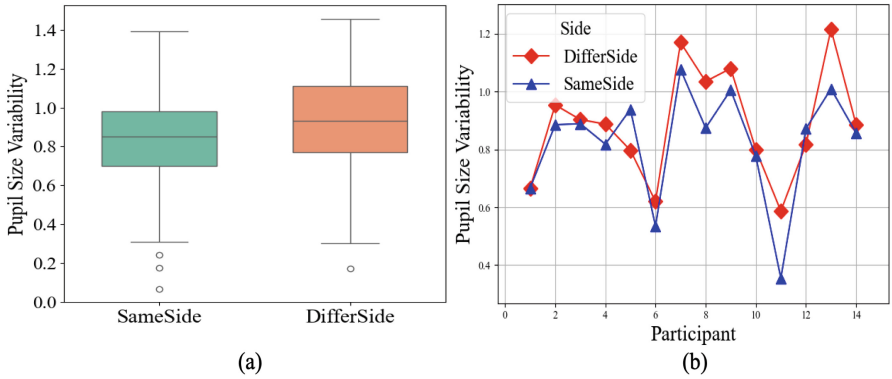

Fig. 2. (a) Comparison of pupil size variability between SameSide and DifferSide conditions. (b) Pupil Size variability across participants in SameSide and DifferSide conditions.

3.2 Analysis of Questionnaire Data

To ensure that participants adhered to the experimental instructions, particularly regarding their attention to the target speaker, we evaluated the accuracy of their responses to questions pertaining to each clip [23]. To achieve this, we employed a partial scoring system based on the Partial Credit Model, a robust framework commonly used in auditory identification tasks [24]. This model enables us to assess varying degrees of correctness in participants' responses.

Our scoring system is categorized as follows:
1. Completely incorrect (0 points): The participant didn't identify any aspect of the target information.
2. Partially correct (0.5 points): The participant correctly identifies either the month or the date.
3. Fully correct (1 point): The participant accurately identifies both the month and the date.

Data Preprocessing. Data recorded from the SameSide and DifferSide groups were analyzed for comparison. To ensure the reliability of the analyzed answers, we excluded 33 out of 192 responses (16 trials $\times$ 12 participants) where participants did not look at the requested face for at least half of the trial time (i.e., visual attention below 50%). For the OnlyListen group, 5 responses were excluded out of 112 (8 trials $\times$ 14 participants) responses where the participants opened their eyes. After data processing, we evaluate the null hypothesis (H0): there is no difference between the groups of answers.

Statistical Analysis. The questionnaire data is independent across groups and sufficiently large to support robust statistical analysis. A normality test (Shapiro-Wilk test) was conducted within each group, indicating that the data follow a non-normal distribution. Given that the data is categorical (with values of 0, 0.5, and 1), non-parametric statistical methods were applied. Specifically, the Mann-Whitney U test and Chi-square test were employed to evaluate the differences across different groups. Both tests showed similar results.

As shown in Table 1, the within-group tests reveal no significant differences, suggesting that these data exhibit comparable levels of accuracy. Consequently, there is insufficient evidence to reject the null hypothesis within groups, leading to the combination of these conditions for subsequent between-group analyses.

Results indicate significant differences in answer accuracy between groups. Specifically, the SameSide condition significantly rejects the null hypothesis compared to the OnlyListen group, suggesting that the SameSide condition results in better accuracy than the OnlyListen group. Similarly, the comparison between SameSide and DifferSide also rejects the null hypothesis. However, no statistically significant difference is found between the "DifferSide" and "OnlyListen" groups, thus failing to reject the null hypothesis in this comparison. These findings highlight the importance of synchronized visual and auditory attention in enhancing information comprehension, particularly emphasizing the contribution of visual cues to speech perception.

The accuracy of the questionnaire responses shows a notable decline across the SameSide, DifferSide, and OnlyListen groups for 10 out of 12 participants. Specifically, the average accuracy declines from 69% in the SameSide group to 56% in the DifferSide group, and 54% in the OnlyListen group. These results indicate that synchronized visual and auditory attention, along with visual cues,

Table 1. Statistical significance test results of questionnaire data within and between groups

Group Type	Within Groups			Between Groups		
	SameSide (LL&RR)	Differside (LR&RL)	OnlyListen (NR&NL)	SameSide & OnlyListen	SameSide & DifferSide	DifferSide & OnlyListen
Significance	n.s.	n.s.	n.s.	$p < 0.001$	$p < 0.05$	n.s.

[a] LL, RR, LR, RL, SameSide, Differside, and OnlyListem represent conditions as detailed in Sect. 2.2.
[b] n.s.: $p > 0.05$, not statistically significant

significantly help the participants to process auditory information, while mismatching visual cues help no more than the OnlyListen setting.

4 Discussion

Our study explored auditory and visual attention mechanisms through various experimental conditions: SameSide, DifferSide, and OnlyListen. The findings reveal that task difficulty significantly influenced participant performance in speech comprehension. Specifically, accuracy was notably higher in the less complex SameSide condition, where visual and auditory information were synchronized, compared to the more challenging DifferSide and OnlyListen conditions.

In the DifferSide condition, where participants had to process auditory stimuli with mismatched facial expressions, they faced significant challenges compared to the SameSide condition with synchronized cues. This condition highlighted the crucial role of matching facial cues in aiding comprehension in complex auditory environments. On the other hand, the OnlyListen condition, which removed visual input, revealed the difficulties participants had in perceiving spatial direction. This condition further underscored the essential role of visual cues in accurately interpreting auditory stimuli.

We also examined the relationship between objective measures of auditory attention and subjective evaluations. Linear regression analysis revealed Pearson correlation coefficients of -0.52, 0.13, and -0.10 for the SameSide, DifferSide, and OnlyListen conditions, respectively. These discrepancies highlight the need for objective standards in evaluating auditory attention, as relying solely on subjective assessments may not fully capture the complexities of auditory processing.

Overall, our findings provide valuable insights into how visual and auditory attention interact, emphasizing the importance of synchronized audiovisual information for effective speech comprehension. These results have significant implications for designing multimodal interfaces and educational strategies, suggesting that incorporating visual cues can enhance auditory attention.

5 Conclusion

The use of AI-generated materials and eye-tracking technology in this study has enhanced our understanding of the interaction between visual and auditory modalities in cognitive processes. Our findings emphasize the crucial role of visual information in auditory attention tasks. Specifically, the increased difficulty observed in the DifferSide condition, where visual and auditory cues were mismatched, underscores the importance of synchronizing visual and auditory attention for effective speech comprehension. Similarly, the OnlyListen condition revealed significant challenges in identifying speakers without visual cues, highlighting the necessity of visual information for accurate speech perception. Future research could investigate diverse populations, such as individuals with hearing impairments, to offer valuable insights into the role of visual and auditory cues under various conditions. Understanding these differences can contribute to the development of more inclusive and accessible multimodal systems.

Acknowledgments. The research is supported by Internal Project of Shenzhen Research Institute of Big Data (Grant No. T00120220002); Undergraduate Research Award of The Chinese University of Hong Kong, Shenzhen; Shenzhen Science and Technology Research Fund, Fundamental Research Key Project Grant No. JCYJ20220818103001002, and Shenzhen Science and Technology Program Grant No. ZDSYS20230626091302006.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. McDermott, J.H.: The cocktail party problem. Curr. Biol. **19**(22), R1024–R1027 (2009)
2. Ahmed, F., Nidiffer, A.R., Lalor, E.C.: The effect of gaze on EEG measures of multisensory integration in a cocktail party scenario. Front. Hum. Neurosci. **17** (2023)
3. Cai, S., Zhu, H., Schultz, T., Li, H.: EEG-based auditory attention detection in cocktail party environment. APSIPA Trans. Signal Inf. Process. **12**(3) (2023)
4. Marsh, J.E., Campbell, T.A., Vachon, F., Taylor, P.J., Hughes, R.W.: How the deployment of visual attention modulates auditory distraction. Atten. Percept. Psychophys. **82**(1), 350–362 (2020)
5. Kolarik, A.J., Pardhan, S., Moore, B.C.J.: A framework to account for the effects of visual loss on human auditory abilities. Psychol. Rev. **128**(5), 913–935 (2021)
6. Hutmacher, F.: Why is there so much more research on vision than on any other sensory modality? Front. Psychol. **10**, 2246 (2019)
7. Schwartz, J.L., Berthommier, F., Savariaux, C.: Seeing to hear better: evidence for early audio-visual interactions in speech identification. Cognition **93**(2), B69–B78 (2004)
8. Oberauer, K.: Working memory and attention—a conceptual analysis and review. J. Cogn. **2**(1), Article 36 (2019)

9. Duncan, J., Martens, S., Ward, R.: Restricted attentional capacity within but not between sensory modalities. Nature **387**(6635), 808–810 (1997)
10. Evans, K.K.: The role of selective attention in cross-modal interactions between auditory and visual features. Cognition **196**, 104119 (2020)
11. Tiippana, K.: What is the McGurk effect? Front. Psychol. **5**, 725 (2014)
12. Gehmacher, Q., et al.: Eye movements track prioritized auditory features in selective attention to natural speech. Nat. Commun. **15**, 3692 (2024)
13. Algazi, V., Duda, R., Thompson, D., Avendano, C.: The CIPIC HRTF database. In: Proceedings of the 2001 IEEE Workshop on the Applications of Signal Processing to Audio and Acoustics (Cat. No.01TH8575), pp. 99–102 (2001)
14. Tobii AB: Tobii Pro Glasses 3 User Manual (2022). https://go.tobii.com/tobii-pro-glasses-3-user-manual
15. Parthasarathy, A., Hancock, K.E., Bennett, K., DeGruttola, V., Polley, D.B.: Bottom-up and top-down neural signatures of disordered multi-talker speech perception in adults with normal hearing. eLife **9**, e51419 (2020)
16. Bradski, G.: The OpenCV library. Dr. Dobb's J. Softw. Tools (2000)
17. Pedrotti, M., et al.: Automatic stress classification with pupil diameter analysis. Int. J. Hum.-Comput. Interact. **30**(3), 220–236 (2014). https://doi.org/10.1080/10447318.2013.848320
18. Kang, O.E., Huffer, K.E., Wheatley, T.P.: Pupil dilation dynamics track attention to high-level information. PLoS ONE **9**(8), e102463 (2014)
19. Minadakis, G., Lohan, K.: Using pupil diameter to measure cognitive load (2018)
20. Krejtz, K., Duchowski, A.T., Niedzielska, A., Biele, C., Krejtz, I.: Eye tracking cognitive load using pupil diameter and microsaccades with fixed gaze. PLoS ONE **13**(9), e0203629 (2018)
21. Smallwood, J., et al.: Pupillometric evidence for the decoupling of attention from perceptual input during offline thought. PloS One **6**(3), e18298 (2011)
22. Beatty, J.: Task-evoked pupillary responses, processing load, and the structure of processing resources. Psychol. Bull. **91 2**, 276–92 (1982). https://api.semanticscholar.org/CorpusID:38397659
23. O'Sullivan, J.A., et al.: Attentional selection in a cocktail party environment can be decoded from single-trial EEG. Cereb. Cortex **25**(7), 1697–1706 (2015)
24. Alain, C., Arnott, S., Picton, T.: Bottom-up and top-down influences on auditory scene analysis: Evidence from event-related brain potentials. J. Exp. Psychol.: Hum. Percept. Perform. **27**, 1072–1089 (2001)